

Brownian motion and stochastic integrals,
viewed from
Ito's isometry and Girsanov's theorem
[DRAFT. Thoughts only]

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There are two dual approaches to this that I've observed. One concerns Stochastic processes, the other — distributions. In this article I try to carry out a pedagogical presentation of material studied at uni, i.e. in a way that feels as “natural” as possible rather than forced.

1 Ultra Basics

1.1 Brownian motion

Brownian motion is defined either

- as a stochastic process $(W_t)_{t \geq 0}$ satisfying three properties:
 1. $W_0 = 0$
 2. W_t is normally distributed with variance t ; and all increments (of the same length) $\Delta W_{s,t}$ share that same distribution;
Later I will write ΔW_t to mean $\Delta W_{0,t}$, so just W_t , but written in the increment style to stress that we are interested in increments of a particular length. By stationarity, expected values then are the same for increments not started at zero, too.
 3. For $s < t$, $W_s \perp \Delta W_{s,t}$, i.e. independent increments
- as a limiting distribution on the space of continuous functions $C(\Omega, \mathbb{R}^n)$ of the distributions of random walks on uniform grid $\{t_i := i/n\}_{i=0}^n$ with jump $1/\sqrt{n}$.

The first view explains how W_t is moving in space, the second — why the variance is $1/t$, i.e. why deviation is $1/\sqrt{t}$.

1.2 Stochastic integral for simple functions

In a Lebesgue-Stieltjes integral style we can integrate simple predictable functions against the brownian motion.

Remember that the Lebesgue-Stieltjes integral against a function (of bounded variation) g ,

$$\int_X f dg$$

works similarly to the standard Lebesgue integral but instead of using a *measure* for the floor increments $(\mu(x - y))$ when infinitesimally multiplying the integrated function, it uses the provided function g 's increment, $\Delta g = g(x) - g(y)$, to “measure” it.

A simple process is a process taking finitely many values that we denote $f_1, \dots, f_k$, i.e. it is expressible as a finite sum

$$f^\omega(t) = \sum_n f_n^\omega \mathbf{1}_{[t_n, t_{n+1}]}(t)$$

So we define the stochastic integral against the brownian motion dW_s of simple predictable functions as

$$\int_0^t f(s) dW_s := \sum_n f_n \Delta W_n$$

2 Square of Δ output is expected to be the Δ input

There are a range of facts that I summarize under this section's title. The chaos of the material is vastly reduced by viewing many results as an instance of this pattern.

2.1 Of W_t itself

2.1.1 Discretely

The second view of Brownian motion above also tells us that considering one jump of the random walk, say

$$\begin{aligned} \text{time: } t_i &\rightarrow t_{i+1} \\ \text{space: } \xi_i &\rightarrow \xi_{i+1}. \end{aligned}$$

With the choice $\Delta t_i = 1/n$, $\Delta \xi_i$ is either $\frac{1}{\sqrt{n}}$ or $-\frac{1}{\sqrt{n}}$. Thus the square $\Delta \xi_i^2$ is most definitely $\frac{1}{n}$. That is,

$$\Delta \xi_i^2 = \Delta t_t$$

— the square of the output's increment equals the input increment.

2.1.2 Continuously

In the limiting distribution $\Delta W_{s,t}$ is definitely does not square to just $(t-s)$, but it is interesting that its mean *does*:

$$\mathbb{E}(\Delta W_t^2) = t.$$

We see this from the property characterising the distribution of the (non-squared) increment ΔW_t :

$$\begin{aligned}\mathbb{E}(\Delta W_t^2) &\equiv \int_{\Omega} \Delta W_t^2 d\mathbb{P} = \int_{\mathbb{R}} y^2 d\mathbb{P}_Z \\ &= \int_{\mathbb{R}} y^2 \Phi(y) dy \stackrel{!}{=} \sigma^2 \int_{\mathbb{R}} \Phi(y) dy = t\end{aligned}$$

where in $\stackrel{!}{=}$ we used (σ^2 denoting the variance of the normal variable, i.e. here just t ; also omitting integral signs)

$$\begin{aligned}y^2 e^{-y^2/2\sigma^2} dy &= -2\sigma^2 y e^{-y^2/2\sigma^2} dy / -2\sigma^2 \\ &= -y de^{-y^2/2\sigma^2} \\ &= -y e^{-y^2/2\sigma^2} + e^{-y^2/2\sigma^2} dy.\end{aligned}$$

Thoughts

Note::::: I can't quite explain yet why in the discrete case randomness goes away ($\Delta \xi_i^2$ is not random), while in the continuous – not (ΔW_t^2 is still random).

2.2 Of the Ito integral for simple functions

Above we computed the expected square of the raw increments ΔW_t . In a martingale-transformation fashion we can consider the “transformed” increments $f\Delta W$ and their integral's expected square:

$$\begin{aligned}\mathbb{E} \left[\left(\int_0^t f(s) dW_s \right)^2 \right] &\equiv \mathbb{E} \left[\left(\sum_n f_n \Delta W_n \right)^2 \right] \\ &\stackrel{\text{lin}}{=} \mathbb{E} \left[\sum_n f_n^2 \Delta W_n^2 \right] + \mathbb{E} \left[\sum_{n \neq m} f_n f_m \Delta W_n \Delta W_m \right] \\ &\stackrel{\perp}{=} \sum_n \mathbb{E} [f_n^2] \underbrace{\mathbb{E} [\Delta W_n^2]}_{\Delta t} + 2 \sum_{n < m} \mathbb{E} [f_n f_m \Delta W_n] \underbrace{\mathbb{E} [\Delta W_m]}_0 \\ &= \mathbb{E} \left[\sum_n f_n^2 \Delta t \right] \\ &\equiv_{\omega} \mathbb{E} \left[\int_0^t f_{\omega}^2(s) ds \right]\end{aligned}$$

which I roughly interpret as

The expected **square** of the modified (by f_t) output (of ΔW_t) equals the **linear** integral of the output's modification's square.

In a way, the ΔW_t^2 collapses inside the integral to Δt , while the f^2 multiple remains as a square.

This peculiar fact carries the name “Ito’s Isometry” (for the very special case of simple functions).